

TCAD Exercise (Team Wise)

Perform all work inside a folder named `ec2l104-tcad`, which is created inside a folder having your name (any one of the two members of the team).

E1. Structure.

Open the SDE of 65 nm node STI n-MOSFET described in Module 2. Run the first node. Save the `.cmd` file and a Visual screenshot of `DopingConcentration` with the region boundaries visible. State the vertex count from the log.

E2. Verify the specification.

Take a horizontal cutline 8 nm below the surface and a vertical cutline through the middle of the channel. Export both. Report the channel length, peak channel doping, and compare each against Figure 1.

E3. Deeper Equilibrium Analysis.

Take a horizontal cutline 8 nm below the surface and observe: `eDensity`, `hDensity`, `Potential`, `SpaceCharge`, `ElectricField`, `eMobility/hMobility`.

E4. Mesh study.

Run the `IdVg` node three times: (a) without the interface refinement, (b) with the interface refinement at 1 nm, and (c) with the interface refinement at 0.3 nm. Plot the three transfer curves on one axis and report the change in I_D at $V_{GS} = 3$ V as a percentage. State which mesh you consider converged and why.

E5. Transfer characteristic.

Run `IdVg` at $V_{DS} = 0.05$ V and at $V_{DS} = 0.4$ V. Export this data from the lab PC to your laptop and use Python to plot both on linear and on logarithmic axes. Extract V_{TH} by linear extrapolation from each, and extract the subthreshold slope.

E6. Output characteristic.

Add `Vgs` as a parameter in the Workbench and produce an I_D - V_{DS} family at four gate voltages (0.3, 0.6, 0.9 and 1.5 V). Export this data from the lab PC to your laptop and use Python to plot I_D - V_{DS} . Also compute g_d with `np.gradient` and report $r_o = 1/g_d$ in saturation at the highest V_{GS} , in $k\Omega \cdot \mu m$.

E7. Physics check.

Re-run the transfer characteristic twice: once with `Enormal` removed, and once with `HighFieldSaturation` removed. Overlay the three curves and explain in three sentences what each model was doing.

E8. Breakdown.

Run the Breakdown node. Report the drain voltage at which I_D reaches 10^{-6} A/ μm . Then produce a plot of `ImpactIonization` on the structure at the last converged bias, and mark where the device is breaking down.

E9. Bulk Terminal Impact on Breakdown.

Add a bulk contact along the bottom face of the substrate in SDE, add the matching `Electrode` entry at 0 V, and repeat E8. Report the new breakdown voltage.

E10. Parameter Variation (A subset of PVT analysis).

- (1) Change the value of gate oxide thickness. The value shall be obtained from the below formula by substituting the last two digits of the roll number of any one of the two group members. Tabulate the entries below. Explain each variation as to why the parameter drifted in the direction observed.

$$t_{ox}(n) = 2.5 + 7.5 \times \frac{n-1}{52} \text{ nm}$$

e.g. $n=01 \rightarrow 2.50 \text{ nm}$; $n=27 \rightarrow 6.25 \text{ nm}$; $n=53 \rightarrow 10.00 \text{ nm}$

t_{ox}	$I_{D,on}$ (@ $V_{DS}=0.4\text{V}$ & $V_{GS}=1.5\text{V}$)	$I_{D,off}$ (@ $V_{DS}=0.4\text{V}$ and $V_{GS}=0\text{V}$)	V_{TH} (V)	$V_{breakdown}$ (V)
5 nm (default)				
Enter your value	0.29mA			

- (2) Change the value of fixed oxide charge at the SiO_2/Si interface. The value shall be obtained from the below formula by substituting the last two digits of the roll number of any one of the two group members. Tabulate the entries below. Explain each variation as to why the parameter drifted in the direction observed.

$$Q_f(n) = 10^{10+0.69897 \times \frac{n-1}{52}} \text{ cm}^{-2}$$

e.g. $n=01 \rightarrow 1.00 \times 10^{10}$; $n=27 \rightarrow 7.07 \times 10^{10}$; $n=53 \rightarrow 5.00 \times 10^{11}$

Q_f	$I_{D,on}$ (@ $V_{DS}=0.4\text{V}$ & $V_{GS}=1.5\text{V}$)	$I_{D,off}$ (@ $V_{DS}=0.4\text{V}$ and $V_{GS}=0\text{V}$)	V_{TH} (V)	$V_{breakdown}$ (V)
0 (default)				
Enter your value				

- (3) For the above 2 questions, separately plot the I_D - V_{DS} and I_D - V_{GS} characteristics superimposing the variations in parameter using Python. Commit the directory in local git and push it into the GitHub repository. This (the git and GitHub repositories and python script) shall be presented also during your viva voce from which questions shall be asked to check your understanding level. Both members of the team must be having this in their laptop.